Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/13/2016
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	09/11/2012	**End Date:**	09/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2017 per F. Hernandez/ARC (Ed., 1/18/17) NOTE: End date is now 12/31/2016, per NSSC information (Ed., 6/15/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX12AO65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 9/28/2012: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants"--grant # NNX10AM86G, per PI move from Miami-Ohio to University of Mississippi. See that project for previous reports] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We plan to study the effects of gravity and light on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	Due to their sessile nature, plants utilize environmental cues to grow and respond to their surroundings. These external cues, taken in concert, direct the growth and development of plants. Two of these cues, light and gravity, play a substantial role in plant orientation and directed growth movements (tropisms). However, very little is currently known about the interaction between light (phototropic) and gravity (gravitropic) mediated growth responses. Utilizing the European Modular Cultivation System (EMCS) on board the International Space Station (ISS), we investigated the interaction between phototropic and gravitropic responses in three Arabidopsis thaliana genotypes, Lansberg wild type as well as mutants of phytochrome A and phytochrome B. Onboard centrifuges were used to create a fractional gravity gradient ranging from reduced gravity up to 1 g. A novel positive blue-light phototropic response of roots was observed during conditions of microgravity, and this response was attenuated at 0.1 g. In addition, a red-light pretreatment of plants enhanced the magnitude of positive phototropic curvature of roots in response to blue illumination. Additionally, a positive phototropic response of roots was observed when exposed to red light, and a decrease in response was gradual and correlated with the increase in gravity. The positive red-light phototropic curvature of hypocotyls when exposed to red light was also confirmed. Both red-light and blue-light phototropic responses were also shown to be a product of directional light intensity. To our knowledge, this is the first characterization of a positive blue-light phototropic response in Arabidopsis roots, as well as the description of the relationship between these phototropic responses in fractional or reduced gravity.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kiss JZ, Vandenbrink JP, Edelmann RE, Medina FJ, Herranz R. "New mechanisms of phototropism are revealed from experiments using the European Modular Cultivation System on the ISS." 22nd European Low Gravity Association symposium, Corfu, Greece, September 29th – October 1st, 2015. 22nd European Low Gravity Association symposium, Corfu, Greece, September 29th – October 1st, 2015. Abstract Book, p. 79. http://www.elgra2015.gr/elgra2015-program-abstracts-book/ , Sep-2015
Abstracts for Journals and Proceedings	Kiss JZ, Vandenbrink JP, Edelmann RE, Herranz R, Medina FJ. "A novel phototropic response observed during experiments utilizing the European Modular Cultivation System." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Abstract Book. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Vandenbrink JP, Dauzart AJ, Kiss JZ. "The effects of clinorotation on plant-microbe symbiosis." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Abstract Book. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Leaving orbit: notes from the last days of American spaceflight.' By M.L. Dean." Choice. 2015 Aug;53:1254. , Aug-2015
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Mission control: inventing the groundwork of spaceflight.' By M.P. Johnson." Choice. 2016 Apr;53:3496. , Apr-2016
Articles in Peer-reviewed Journals	Johnson CM, Subramanian A, Edelmann RE, Kiss JZ. "Morphometric analyses of petioles of seedlings grown in a spaceflight experiment." Journal of Plant Research. 2015 Nov;128(6):1007-16. Epub 2015 Sep 16. http://dx.doi.org/10.1007/s10265-015-0749-0 ; PubMed PMID: 26376793 , Nov-2015

Articles in Peer-reviewed Journals	Vandenbrink JP, Kiss JZ. "Space, the final frontier: a critical review of recent experiments performed in microgravity." Plant Science. 2016 Feb;243:115-9. Epub 2015 Nov 7. Review. http://dx.doi.org/10.1016/j.plantsci.2015.11.004 ; PubMed PMID: 26795156 , Feb-2016
Articles in Peer-reviewed Journals	Dauzart AJC, Vandenbrink JP, Kiss JZ. "The effects of clinorotation on the host plant, Medicago truncatula, and its microbial symbionts." Frontiers in Astronomy and Space Sciences. 2016;3. Published on 26 February 2016. http://dx.doi.org/10.3389/fspas.2016.00003 , Feb-2016
Awards	Kiss JZ. (John Z. Kiss) "Outstanding Reviewer for 'consistently and expeditiously delivered comprehensive, discerning reviews to the Journal's authors.' National Research Council of Canada, April 2016." Apr-2016